

Setup Protocols for Sender Anonymity

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Abstract—Anonymous communication is essential for secure and private interactions over public networks. Existing solutions that provide provable anonymity rely on the so-called *simple I/O setting*, where every participant sends and receives the same number of messages, masking their true communication pattern. The only known way to enforce this setting is through *dialing protocols*. Such protocols establish pairwise conversations, but each recipient inevitably learns who attempted to contact them, violating *sender anonymity*, the guaranty that even the recipient cannot determine who attempted to contact them.

In this work, we introduce the notion of *enhanced dialing protocols*, a broad class of protocols that enforce the *simple I/O setting*. We also initiate the first formal study of such protocols with respect to sender anonymity. We introduce a framework that captures three key properties: security, correctness, and fairness. Within this framework, we present *Fusion*, a protocol that achieves perfect correctness and fairness while incurring only unavoidable leakage, and *Fusion+*, a differentially private variant that reduces this leakage at the cost of some correctness. Through theoretical analysis, we quantify the fundamental trade-off between privacy and correctness in *Fusion+*.

1. Introduction

Motivating example. The ability to communicate anonymously over public networks is fundamental to secure communication [1], [2] and underpins many privacy-preserving technologies [3], [4], [5]. Consider the following example: Alice, a whistleblower, wishes to send a story about Company X to the New York Times. Even if she encrypts the message so that only the Times can read it, publicly visible metadata (such as traffic patterns) may still reveal that Alice is communicating with the Times. This alone could be incriminating. Thus, protecting anonymity against adversaries who monitor and analyze network traffic is essential for safe whistleblowing. Moreover, it may be desirable that even the New York Times does not learn Alice’s identity, so that it cannot reveal it, even under legal pressure.

The demand for anonymous communication is significant, as evidenced by the widespread use of networks

such as Tor [6], which serves millions of users daily,¹ as well as Freenet [7] and Crowds [8]. These systems are designed to obscure the link between senders and recipients while addressing practical concerns such as low latency and efficient bandwidth usage. However, they do not provide strong cryptographic guarantees that make it impossible for a resourceful adversary (e.g., an ISP or nation-state actor with full access to network traffic) to determine who is communicating with whom. As a result, such systems often compromise anonymity and remain vulnerable to attacks, including network timing attacks; see e.g. [9], [10], [11].

Over the past decade, significant progress has been made in the design of provably anonymous channels. These advances include formal definitions [12], [13], [14], [15], [16], [17], [18] and security guarantees against active adversaries (who observe all network traffic and may control up to half of the network) [13], [19], [20], [21], [22], [23], as well as results that account for network churn [18] and asynchronous communication [24].

1.1. Anonymity against the standard adversary

Let us begin by clarifying what we mean by anonymity.

In the cryptographic literature (see, e.g., work on multi-party computation), the “adversary” is modeled as representing perfect collusion among some of the parties in the network. Three standard adversary models are typically considered, in increasing order of power: the *network adversary*, which observes all network traffic; the *passive adversary*, which additionally corrupts some parties and observes their internal state, including messages sent to and from those parties; and the *active adversary*, which further controls the corrupted parties and may cause them to deviate arbitrarily from the protocol. When there are corrupted parties, the adversary’s view consists of the combined observations of all corrupted parties together with complete access to network traffic on all links at all times. In this paper, we consider the strongest of these models, namely the active adversary.

Anonymity is defined relative to the information available to such an adversary. Anonymity requires that the adversary cannot distinguish between any two admissible

1. Data retrieved from <https://metrics.torproject.org/> on January 30, 2026.

options for the communication pattern: for example, between the case where Alice sends a message to Bob and the alternative where Alexander (not Alice) sends a message to Bob. In some settings, we may also wish to protect the sender’s identity even from the recipient (e.g., the New York Times); we refer to this stronger property as *sender anonymity*.

In this paper, we focus on the sender anonymity of peer-to-peer (P2P) communication protocols. Formally, we model such a protocol with a fixed set of participants $P_1, P_2, \dots, P_N$. Each participant is given an input specifying a set of message-recipient pairs. Anonymity is achieved if the adversary cannot distinguish between any two possible input vectors, where each vector corresponds to a “scenario” defined by the inputs of all participants.

1.2. Overcoming the Simple I/O Restriction

It is known how to achieve P2P sender anonymity against an active adversary controlling a constant fraction of the parties, but only in fairly restrictive settings (see, for example, Protocols Π_p [13], Pi-Tree (Π_Δ) [14], and Pi-Butterfly ($\Pi_\times$) [14]). In particular, these works rely on the so-called *simple I/O setting*, which imposes an unnatural restriction on participants’ communication patterns by requiring each participant to send exactly one message and receive exactly one message in return. To gain intuition for why such a restriction is helpful, consider the following two scenarios.

- **Scenario 1:** Alice sends a (fixed-length) message to Bob, who sends a (fixed-length) message to Charlie.
- **Scenario 2:** Both Alice and Bob send (fixed-length) messages to Charlie.

The adversary can distinguish between the two scenarios simply by counting how many messages Charlie receives: one versus two. Dummy traffic can provide network cover but cannot strengthen sender anonymity.

The current state of provable sender anonymity can be described as follows. If we decompose the problem into (a) enforcing a restricted communication pattern and (b) achieving sender anonymity assuming such a pattern, then existing work resolves (b), while (a) has remained open; thus, this paper focuses on addressing (a). The existence of *setup protocols* for (a) would therefore imply practical sender anonymity without relying on the unnatural simple I/O restriction.

We study two privacy issues arising from the metadata of point-to-point communication. The first is hiding the identity of the session initiator from the responder (part of the adversary), even when the responder observes the entire network transcript; this is the strongest per-session anonymity notion that can be meaningfully required. This captures, for example, the whistleblowing scenario described above. The second is to hide the “popularity” of honest parties, measured by the number of sessions initiated with them. These requirements are in addition to the usual goals of hiding message contents and concealing session endpoints

from a network adversary, which are already well understood. We therefore abstract away these solved issues and focus exclusively on session setup.

Our model routes all connection requests through a bulletin-board functionality that decides which sessions are established. (We emphasize that this abstraction captures both centralized and distributed implementations, as discussed later.) Ensuring initiator anonymity in this model is relatively straightforward, using standard pseudonymous techniques. In contrast, protecting popularity information is more subtle: bounding the number of concurrent sessions per party forces the bulletin board to reject some requests, and this action may itself leak information. To address this, we use differential privacy in the shuffle model to trade off popularity leakage against the rate of rejected connection attempts.

1.3. Dialing Protocols

The challenges above manifest most clearly in the design of dialing protocols (described below). While it is easy to control the number of messages sent by each participant, ensuring that each recipient receives a specific number of messages is much more difficult. The challenge arises because parties cannot easily coordinate the recipients without compromising anonymity. For instance, Alice and Alexander cannot safely disclose that they both intend to send a message to Bob. Therefore, a central open problem in this area is how to conceal the total number of messages directed to each recipient.

Vuvuzela [19] introduced the now standard dichotomy between a dialing protocol and a conversation protocol. The *dialing protocol* is responsible for establishing peer-to-peer conversations, while the *conversation protocol* enables the actual exchange of messages once the connections are made.

Dialing protocols are the primary mechanism used to enforce simple I/O constraints. As an example, in Vuvuzela, suppose Alice wants to communicate with Bob. She begins by sending an invitation to Bob’s mailbox to see if he is willing to engage. Each participant’s mailbox is shared among multiple users, helping to obscure the volume of invitations that each party receives. Once all invitations are collected, each participant downloads the contents of their mailbox and initiates the conversation protocol with the party whose invitation they choose to accept. Importantly, all invitations are encrypted, ensuring that only the intended recipient can read them.

Beyond Vuvuzela, other dialing protocols have been proposed [25], [26], [27], [28], [29], [30], [31], each introducing routing mechanisms to hide who sent which invitation.

Thus, while the only known mechanism for enforcing the simple I/O setting is through dialing protocols [19], [25], [26], [27], [28], [29], [30], [31], which securely implements a restricted form where Alice sends to Bob if and only if Bob sends to Alice; this restriction limits both expressiveness and introduces anonymity leaks. First, as mentioned by Angel and Setty [32], none of these provides sender anonymity as each recipient learns the entire set of parties

who wish to talk to him. This is problematic in scenarios such as the whistleblowing example above. Second, these protocols typically assume a communication pattern where pairs of parties exchange messages [33]; in other words, Alice cannot send a message to Bob while Bob sends a message to any other party but Alice (e.g., Charlie).

1.4. Our Contributions

Inspired by prior work, we adopt a broad definition of an *enhanced* dialing protocol as any approach that ensures each participant sends exactly d fixed-length messages and is slated to receive exactly d messages in return (enforcing a generalized simple I/O setting; see Figure 1). Despite be-

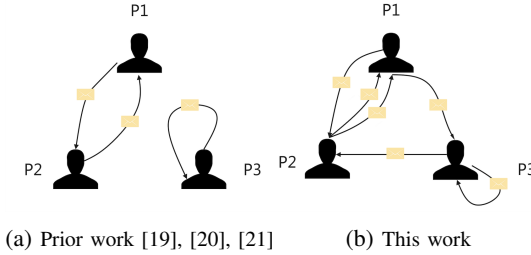


Figure 1: Comparison of communication pattern settings between existing dialing protocols (a) and our enhanced dialing protocol (b). Existing dialing protocols (a) only support a restricted simple I/O settings, where each participant can send and receive one message. In contrast, our proposed dialing protocol (b) enables a more general simple I/O setting in which each participant can send multiple messages, and the number of messages sent and received by each participant is equal.

ing a fundamental building block for anonymous channels, the necessary and sufficient conditions for achieving such protocols have not been previously studied.

We undertake a systematic study of enhanced dialing protocols with the explicit goal of achieving sender anonymity. Our main contributions are as follows:

- We establish a lower bound showing that if an enhanced dialing protocol aims for perfect correctness and fairness, some degree of information leakage is inevitable.
- We present the first enhanced dialing protocol that guarantees private sender anonymity while maintaining fairness and an acceptable level of correctness.

Roadmap of the paper. We first define our problem setting, including the inputs, outputs, and underlying assumptions. Then, in Sections 3.1-3.3, we formalize three core properties that an enhanced dialing protocol for sender anonymity should satisfy:

- 1) **Security.** We provide two security definitions: one based on indistinguishability (Definition 2) and one based on differential privacy (Definition 3). The indistinguishability-based definition uses a leakage function L , specifying that a secure enhanced dialing protocol Σ reveals no information beyond L .
- 2) **Correctness.** This metric quantifies the protocol’s effectiveness in maximizing the acceptance rate of invitations by their intended recipients (Definition 4).

- 3) **Fairness.** Fairness ensures that when multiple invitations target the same recipient, each invitation has an equal probability of being accepted (Definition 5).

In Section 4, we prove that any enhanced dialing protocol that is both perfectly correct and fair cannot satisfy differential privacy (Corollary 1). We further characterize the inherent information leakage of such protocols: any perfectly correct and fair protocol may reveal the *popularity pattern*, the number of invitations from honest parties for each recipient (Theorem 3).² Additionally, we show that the simple approach of anonymously posting the identities of intended communication partners is optimal, leaking at most the popularity pattern (Theorem 2). This method forms the basis of our Fusion protocol.

While our construction uses a single server in charge of a bulletin board, our lower bounds apply broadly to any perfectly correct and fair enhanced dialing protocol, including those that employ advanced cryptographic techniques (e.g., multiparty computation) or distribute computation across multiple servers. *Note:* For simplicity, we will refer to the combination of the server and bulletin board as simply the bulletin board.

In Section 4.3.1, we complement our theoretical results with experiments, demonstrating that the leaked information can be efficiently reconstructed. The recovery depends on the fraction of corrupted participants and the number of invitations per round, rather than the total network size. For example, in a network of approximately 50,000 users where each sends $d = 20$ invitations per round and the adversary controls 3% of participants, the popularity pattern can be recovered with 90% accuracy after 20 rounds (Figure 4).

Finally, to mitigate this inherent leakage (at the cost of some correctness), we design Fusion+, a differentially private variant of Fusion. Leveraging an anonymous posting protocol, Fusion+ can be analyzed in the shuffle model of differential privacy, minimizing the noise required for strong privacy guarantees. We prove that Fusion+ satisfies (ϵ, δ) -differential privacy (Theorem 4) while maintaining fairness, and we quantitatively evaluate the privacy-correctness trade-off. For instance, in a network of 50,000 users, Fusion+ achieves $(\epsilon = \ln 3, \delta = 10^{-4})$ -differential privacy and $(\alpha = 0.739, \beta = 0.05)$ -correctness.

In summary, we present the first provably sender-anonymous (enhanced) dialing protocol. Combined with existing protocols that achieve sender anonymity in the simple I/O setting (e.g., [13], [14]), our work demonstrates that it is possible to achieve practical sender anonymity without imposing unnatural restrictions on the communication pattern.

2. Notation and Building Blocks

In this section, we introduce the notation and fundamental concepts that form the basis for our subsequent theoretical analysis.

2. Corrupted parties’ inputs are also revealed, but this is unavoidable.

2.1. Notation

For an integer n , we denote the set $[n]$ as $\{1, 2, \dots, n\}$. The notation $x \leftarrow \text{Set}$ means that x is sampled uniformly at random from the set, Set; $x \leftarrow \text{Dist}$ means x is sampled according to the distribution, Dist; and $x \leftarrow \text{Algo}$ means x is a random output generated by the algorithm, Algo. A function $\nu : \mathbb{N} \mapsto \mathbb{R}$ is considered negligible if it decreases faster than any inverse polynomial in the security parameter. An event occurs with negligible (resp. with overwhelming) probability if the probability of that event occurring is bounded above (resp. below) by some negligible function (resp. one minus some negligible function).

2.2. Anonymous Posting Protocol

A posting protocol is a protocol in which participants $P_1, \dots, P_N$ post messages to a public bulletin board. We use a *correct* and *anonymous* posting protocol as a building block in our enhanced dialing constructions; to that end, we provide formal definitions for this primitive here. Note that, while dialing protocols are essential for establishing anonymous P2P communication, no such setup is necessary for an anonymous posting protocol where all participants send their messages to a bulletin board.

A posting protocol is correct if (when everyone adheres to the protocol), all the messages get posted on the bulletin board [13, Definition 6].

A posting protocol is anonymous if given any two scenarios $\bar{\sigma}^0$ and $\bar{\sigma}^1$, it is impossible for an adversary to discern whether the protocol was executed in scenario $\bar{\sigma}^0$ or $\bar{\sigma}^1$. As examples of scenarios, in $\bar{\sigma}^0$, Alice posts “hello” and Bob posts “goodbye;” whereas in $\bar{\sigma}^1$, Alice posts “goodbye” and Bob posts “hello.”

The scenarios are defined by the inputs of the parties, which specify the messages they will post. The input σ_i of party P_i consists of a set of messages chosen from the message space (a set of fixed length messages). The inputs $\bar{\sigma}^0 = (\sigma_1^0, \sigma_2^0, \dots, \sigma_N^0)$ and $\bar{\sigma}^1 = (\sigma_1^1, \sigma_2^1, \dots, \sigma_N^1)$ are chosen such that the inputs of the corrupted parties and the set of everyone’s messages are the same; we denote this as $\bar{\sigma}^0 \equiv_{\text{Bad}} \bar{\sigma}^1$ where Bad is the set of corrupted parties.

The formal definition of anonymity [23, Definition 4] is given with respect to the following game:

Posting game. $\text{PostingGame}(1^\lambda, \Pi, \mathcal{A})$ is parameterized by the security parameter 1^λ , the adversary $\mathcal{A}$, and the posting protocol Π . This game is played between $\mathcal{A}$ and the challenger $\mathcal{C}$.

First, $\mathcal{A}$ and $\mathcal{C}$ set up their keys: $\mathcal{A}$ selects the corrupted parties, denoted as $\text{Bad} \subseteq \{P_1, P_2, \dots, P_N\}$, and informs $\mathcal{C}$ of this choice. Then $\mathcal{C}$ generates the keys for the honest parties according to the protocol Π and sends the public portions of these keys, $\text{pk}(\text{Honest})$, to $\mathcal{A}$. In return, $\mathcal{A}$ submits the public keys of the corrupted parties, $\text{pk}(\text{Bad})$.

Next, $\mathcal{A}$ presents a challenge to $\mathcal{C}$: $\mathcal{A}$ sends two input vectors $\bar{\sigma}^0$ and $\bar{\sigma}^1$ such that $\bar{\sigma}^0 \equiv_{\text{Bad}} \bar{\sigma}^1$. $\mathcal{C}$ then randomly selects a bit $b \leftarrow \{0, 1\}$, and $\mathcal{A}$ and $\mathcal{C}$ interact in a run of

Π using the input $\bar{\sigma}^b$. At the end of the execution of the protocol, $\mathcal{A}$ makes a guess b' for the bit b and wins the game if $b' = b$.

We define anonymity as follows:

Definition 1 (Anonymity [23, Definition 4]). *A posting protocol Π is statistically (resp. computationally) anonymous from the adversary $\mathcal{A}$ if every (resp. every computationally bounded) adversary $\mathcal{A}$ wins $\text{PostingGame}(1^\lambda, \Pi, \mathcal{A})$ with only negligible advantage, i.e., $|\Pr[\mathcal{A} \text{ wins } \text{PostingGame}(1^\lambda, \Pi, \mathcal{A})] - \frac{1}{2}| = \text{negl}(\lambda)$.*

Note: We assume the anonymous posting protocol includes an underlying key generation algorithm KeyGen allowing each party to generate its own public-private key pair.

2.3. Histograms in the Shuffle Model

In the *central model* of differential privacy [34], a trusted curator collects information from participants and publishes noisy or sanitized statistics based on these data. This approach differs from the *local model* [35], where there is no trusted curator; instead, each participant is responsible for protecting their own data privacy. Typically, the local model results in a considerably worse privacy-utility trade-off compared to the central model; see, e.g., [36].

To address this issue, a middle-ground approach known as the *shuffle model* has emerged in recent years [37], [38]. In this model, the curator is untrusted (similar to the local model), but receives data from participants in a way that prevents them from knowing which data belongs to which individual.

Balle, Bell, Gascón, and Nissim [39] introduced the “single-message shuffle model” protocols. In this approach, each user applies a local randomizer to their data, producing one or more noisy messages to send to a trusted shuffler. The shuffler then anonymizes the collected data by randomly permuting all the messages before releasing them to the untrusted curator. This shuffling process is particularly effective for computing some types of statistics because, without relying on a trusted curator, anonymizing the data significantly reduces the amount of noise required for privacy.

A relevant result for this paper pertains to the release of a sanitized histogram. Suppose that there are n users, each with input $x_i \in [k]$. User i applies a local randomizer $\mathcal{R}$ to produce a single message y_i . The multi-set $\{y_1, \dots, y_n\}$ is then permuted by a uniformly random shuffle σ into $(y_{\sigma(1)}, \dots, y_{\sigma(n)})$, and the curator computes the histogram of the shuffled messages. Balle et al. proved:

Theorem 1 (Shuffled-model privacy [39, Theorem 3.1]). *The shuffled mechanism satisfies (ϵ, δ) -differential privacy for any $k, n \in \mathbb{N}$, $\epsilon \leq 1$ and $0 < \delta \leq 1$, provided that*

$$p = \max \left\{ \frac{14k \log(2/\delta)}{(n-1)\epsilon^2}, \frac{27k}{(n-1)\epsilon} \right\} < 1,$$

where p is the randomizer’s noise-injection probability.

We use this result to analyze our construction in Section 5, in which the anonymous posting protocol acts as the shuffler, and the public bulletin board acts as the untrusted analyzer.

3. Modeling the Problem

Here, we describe the technical problem we wish to solve. In many anonymity protocols, the objective is to facilitate private communication between pairs of users who have mutually agreed to interact. These protocols often assume that communication patterns (that is, who is communicating with whom) have been established in advance. A dialing protocol serves as a mechanism for creating and determining these communication relationships.

There are N participants, denoted as $\mathcal{P} \stackrel{\text{def}}{=} \{P_1, P_2, \dots, P_N\}$. Each participant P_i wishes to communicate with up to d other participants. These desired connections are treated as inputs to the dialing protocol. At the end of the dialing protocol, each participant P_i will return the d parties to whom they will send a message in the anonymity protocol.

More formally, the input $X_i = X_{i,1}, \dots, X_{i,d}$ for participant P_i in the dialing protocol is a multi-set (allowing repetitions) of the recipients of d messages that P_i wishes to send. By the input vector, we mean the vector $\vec{X} = (X_1, X_2, \dots, X_N)$, containing the inputs of all the participants. We sometimes refer to the input of an individual party as the “party’s invitee(s).”

The output $O_i = \{O_{i,1}, O_{i,2}, \dots, O_{i,d}\}$ for participant P_i in the dialing protocol is a multi-set of the recipients of the d messages that P_i is permitted to send (in the anonymity protocol). By the output vector, we mean the vector $\vec{O} = (O_1, O_2, \dots, O_N)$, which contains the outputs of all the participants.

In our problem setting, we consider a bulletin board as a medium where participants can submit their invitations. We also make use of a correct and anonymous posting protocol (defined in Section 2) that operates within an appropriate communication model. Furthermore, we rely on a public key infrastructure (PKI) to facilitate key distribution; specifically, we assume that the public key of each party, including that of the bulletin board, is publicly accessible.

The goal of the enhanced dialing protocol is to establish communication relationships in the “generalized” simple I/O setting (i.e., every participant is represented exactly d times in the outputs) while ensuring security, correctness, and fairness, described succinctly below, and formally defined in Sections 3.1-3.3:

- **Security.** The server and corrupted participants learn minimal information about the inputs of honest parties, even with collusion.
- **Correctness.** The protocol maximizes the overlap between each participant’s input (who they want to talk to) and output (who they get to talk to).
- **Fairness.** The protocol guarantees that all participants have an equal opportunity to connect with their desired recipients.

In this work, we focus on the enhanced dialing protocols that achieve these properties simultaneously.

3.1. Security

In this section, we formally define what it means for an enhanced dialing protocol to be secure.

3.1.1. Adversary Model

We assume that the bulletin board and some of the participants are corrupted. Together, they comprise the standard *active adversary* who controls the corrupted parties (causing them to behave in arbitrary ways, including deviating from the protocol) and, additionally, monitors the traffic on all communication links.

3.1.2. Leakage-based Security

Recall that the input to the dialing protocol consists of a vector of inputs $(X_1, X_2, \dots, X_N)$, where each input X_i indicates with whom party P_i wishes to communicate. For the enhanced dialing protocol to be considered secure, the adversary should learn minimal information about the inputs of the honest parties. In other words, the adversary should not be able to distinguish whether the enhanced dialing protocol was executed with input $\vec{X}^0$ or, alternatively, with input $\vec{X}^1$.

It is important to note that there are necessary restrictions on the adversary’s selection of $\vec{X}^0$ and $\vec{X}^1$. As a concrete example, consider the scenario in which the party P_j is corrupted. The adversary can trivially distinguish between any pair of input vectors $\vec{X}^0 = (X_1^0, X_2^0, \dots, X_N^0)$ and $\vec{X}^1 = (X_1^1, X_2^1, \dots, X_N^1)$ where $X_j^0 \neq X_j^1$. Thus, $\vec{X}^0$ and $\vec{X}^1$ must be the same in the inputs of the corrupted parties.

Restrictions placed on the inputs of honest parties are “leakages that are allowed” by the security definition. For example, suppose that we define security to mean that the adversary cannot distinguish between $\vec{X}^0$ and $\vec{X}^1$ as long as $\bigcup_{i=1}^N X_i^0 = \bigcup_{i=1}^N X_i^1 =$ (some fixed set of messages). Then, a protocol that satisfies this definition does not necessarily prevent the adversary from learning the total set of invitations.

More generally, we can define leakage as a function, L , of the inputs as follows:

Let $\mathcal{X}$ be the set of all possible input vectors. Fixing the corrupted parties Bad and the leakage function L , we partition $\mathcal{X}$ into equivalence classes, $\mathcal{X}_1, \mathcal{X}_2, \dots$, where each class $\mathcal{X}_i$ corresponds to a leakage value ℓ_i and is the set of input vectors $\vec{X}$ such that $L(\vec{X}) = \ell_i$. If two input vectors $\vec{X}$ and $\vec{X}'$ fall into the same equivalence class, we write this as $\vec{X} \equiv_{\text{Bad}, L} \vec{X}'$.

We propose a game-based definition of security of a dialing protocol using the following game.

Dialing game. $\text{DialingGame}(1^\lambda, \Sigma, \mathcal{A}, L)$ is parametrized by the security parameter 1^λ , the dialing protocol Σ , the adversary $\mathcal{A}$, and the leakage function L .

First, the adversary $\mathcal{A}$ and the challenger $\mathcal{C}$ set up the parties’ keys: $\mathcal{A}$ chooses a subset $\text{Bad} \subseteq \{P_1, P_2, \dots, P_N\}$

of the parties to corrupt and sends this information to the challenger $\mathcal{C}$. For each honest party, $\mathcal{C}$ generates a key pair for the party as specified by Σ ; the public keys of the honest parties, $\text{pk}(\text{Honest})$, are sent to $\mathcal{A}$. $\mathcal{A}$ sends the public keys of the corrupted parties, $\text{pk}(\text{Bad})$, to $\mathcal{C}$.

Next, the input is selected: $\mathcal{A}$ picks two input vectors $\vec{X}^0$ and $\vec{X}^1 \equiv_{\text{Bad}, L} \vec{X}^0$ from the same equivalence class w.r.t. Bad and L and sends them to $\mathcal{C}$. $\mathcal{C}$ chooses a random bit $b \leftarrow \{0, 1\}$.

At the end of the execution, $\mathcal{A}$ computes a guess b' for b from its view $V^{\Sigma, \mathcal{A}, \text{Bad}}(1^\lambda, \vec{X}^b)$ and wins the dialing game if $b' = b$. See Figure 2.

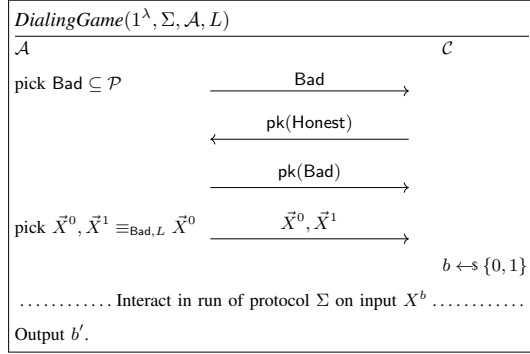


Figure 2: Schematic of the dialing game.

The formal definition of security is given as follow:

Definition 2 (Security). A dialing protocol Σ is statistically (resp. computationally) secure w.r.t. the leakage L if every (resp. every probabilistic polynomial-time) adversary $\mathcal{A}$ wins $\text{DialingGame}(1^\lambda, \Sigma, \mathcal{A}, L)$ with only negligible advantage, i.e.,

$$\left| \Pr[\mathcal{A} \text{ wins } \text{DialingGame}(1^\lambda, \Sigma, \mathcal{A}, L)] - \frac{1}{2} \right| = \text{negl}(\lambda).$$

Note: The security definition above is stated for a single round of the enhanced dialing protocol. If an enhanced dialing protocol is secure without any leakage, then in the multi-round setting the adversary's advantage remains negligible. More generally, if it is secure with respect to a leakage function L , then over multiple runs of the protocol on input vectors $(\vec{X}_1, \dots, \vec{X}_R)$, the total leakage is at most $(L(\vec{X}_1), \dots, L(\vec{X}_R))$.

We will use the above notion of security to analyze our first enhanced dialing protocol, Fusion, in Section 4.

3.1.3. Differential Privacy

In Theorem 3, we will show that enhanced dialing protocols designed to achieve both perfect correctness (we will define this in Definition 4) and fairness (in Definition 5) must leak at least the popularity pattern (which may be unacceptable in most cases).

To address this, in Section 5, we modify Fusion, using a differentially private mechanism. This allows for a relaxation of correctness in a controlled manner that also differentially privately hides what Fusion leaks. To support

this approach, we provide an alternative notion of security based on standard differential privacy [34].

Given two inputs $\vec{X}^0 = (X_1^0, \dots, X_N^0) = (\{X_{1,1}^0, \dots, X_{1,d}^0\}, \dots, \{X_{N,1}^0, \dots, X_{N,d}^0\})$ and $\vec{X}^1 = (X_1^1, \dots, X_N^1) = (\{X_{1,1}^1, \dots, X_{1,d}^1\}, \dots, \{X_{N,1}^1, \dots, X_{N,d}^1\})$, let the distance between these input vectors, $d(\vec{X}^0, \vec{X}^1)$, be given by:

$$d(\vec{X}^0, \vec{X}^1) = \sum_{i=1}^N |X_i^0 \nabla X_i^1|,$$

where (∇) denotes the symmetric difference.

Two input vectors, $\vec{X}^0$ and $\vec{X}^1$, are *neighboring* if $d(\vec{X}^0, \vec{X}^1) \leq 2$.

Definition 3 (Differential privacy). An enhanced dialing protocol Σ satisfies (ϵ, δ) -differential privacy if for every adversary $\mathcal{A}$, every pair $(\vec{X}^0, \vec{X}^1)$ of neighboring inputs differing only on an honest party's input, and every set $\mathcal{V}$ of adversarial views,³ the following holds:

$$\Pr[V^{\Sigma, \mathcal{A}}(\vec{X}^0) \in \mathcal{V}] \leq e^\epsilon \Pr[V^{\Sigma, \mathcal{A}}(\vec{X}^1) \in \mathcal{V}] + \delta.$$

We say that Σ is computationally (ϵ, δ) -differentially private [40] if the above condition holds for all polynomially bounded adversaries, and δ is a negligible function.

Note: The definition of differential privacy above is stated for a single execution of the protocol. Under standard sequential composition, running the protocol for R executions yields $(R\epsilon, R\delta)$ -differential privacy [41].

3.2. Correctness

Let the input vector be $\vec{X} = (X_1, X_2, \dots, X_N)$, and let the output vector be $\vec{O} = (O_1, O_2, \dots, O_N)$ such that X_i and O_i are the input and output of party P_i , respectively.

To make the enhanced dialing protocol as effective as possible, we aim to increase the number of input invitations “realized” in the outputs. In other words, our goal is to maximize the total number of inputs $X_{i,j}$ that uniquely correspond to an output $O_{i,k}$.

For $i, j \in [N]$, let $C_{i \rightarrow j}$ represent the number of occurrences, or the count, of P_j in the input X_i for P_i . This can be

expressed as $C_{i \rightarrow j} = |X_i \cap \overbrace{\{P_j, P_j, \dots, P_j\}}^{d \text{ times}}|$ where $|\cdot|$ denotes the cardinality of the multi-set. Let $C_j = \sum_{i=1}^N C_{i \rightarrow j}$ denote the total count of P_j in $\vec{X}$.

Similarly, for $i, j \in [N]$, we can let $O_{i \rightarrow j}$ represent the number of occurrences of P_j in the output O_i for P_i . It can

be expressed as $O_{i \rightarrow j} = |O_i \cap \overbrace{\{P_j, P_j, \dots, P_j\}}^{d \text{ times}}|$.

For $i, j \in [N]$, define $M_{i \rightarrow j}$ as the number of invitations for P_j in X_i that uniquely match an occurrence of P_j in O_i . In other words, this is the number of P_j 's in the intersection

of X_i and O_i , i.e., $M_{i \rightarrow j} = |X_i \cap O_i \cap \overbrace{\{P_j, P_j, \dots, P_j\}}^{d \text{ times}}|$.

3. By the *adversarial view*, we mean the set of observations that the adversary makes during a run of the protocol.

Thus, $M_j = \sum_{i=1}^N M_{i \rightarrow j}$ represents the total number of invitations for P_j that are “realized” in the outputs.

Definition 4 ((α, β) -correctness). *Let $\alpha, \beta \in [0, 1]$ be correctness parameters.*

Assuming a passive adversary and that all parties adhere to the protocol, an enhanced dialing protocol is (α, β) -correct if the following condition holds:

$$\Pr \left[\frac{\sum_{j=1}^N M_j}{\sum_{j=1}^N \min(C_j, d)} < 1 - \alpha \right] \leq \beta.$$

If $\alpha = \beta = 0$, the enhanced dialing protocol is perfectly correct.

3.3. Fairness

By fairness, we mean that when multiple parties try to communicate with the same recipient, each invitation should have an equal chance of being realized.

Using notation from above, let $C_{i \rightarrow j}$ represent the number of messages that party P_i wishes to send to party P_j (i.e., the number of times P_j appears in their input X_i). Let $C_j = \sum_{i=1}^N C_{i \rightarrow j}$ denote the total number of invitations for P_j , and let $M_j \leq C_j$ be the number of these that are actually realized.

Definition 5 (Fairness). *Assuming a passive adversary and that all parties adhere to the protocol, an enhanced dialing protocol is said to satisfy fairness if it meets the following condition:*

$$\mathbb{E}[M_{i \rightarrow j}] = \frac{C_{i \rightarrow j} \mathbb{E}[M_j]}{C_j}, \quad \forall i, j.$$

Here, the expectation is taken over the internal randomness of the protocol.

Note: The adversary models considered for different properties are not identical. Security guarantees are defined with respect to a colluding active adversary model. In contrast, correctness and fairness are defined assuming a passive adversary and that all parties adhere to the protocol. Intuitively, this means that our protocol preserves privacy even when the bulletin board and some participants are corrupted and behave arbitrarily (and possibly all colluding). When all parties follow the protocol, the dialing mechanism achieves (near-optimal) efficiency in realizing invitations, and the probability that a given invitation is realized depends only on the recipient’s load and not on the identity of the sender.

4. Fusion

Recall the problem described in Section 3. We consider a situation with N participants, denoted as $P_1, P_2, \dots, P_N$, where some participants may be corrupted. Each participant P_i has an input X_i , which consists of a set of d parties or “invitees” with whom P_i wishes to communicate.

We assume that there is a secure and anonymous posting protocol in place with an underlying key generation algorithm KeyGen.

Each participant is associated with a pair of keys generated by KeyGen: a public key pk_i and a secret key sk_i . We also assume the existence of a bulletin board BB that has its own key pair, represented as (pk_{BB}, sk_{BB}) . The public portions of all keys are made available through a public key infrastructure (PKI).

We aim to design an enhanced dialing protocol that is simultaneously secure (with “minimal” information leakage), correct, and fair. Initially, we will focus on solutions that achieve perfect correctness and fairness.

Consider the following simple example with $N = 3$ participants and $d = 1$ invitation per party. The inputs for the three participants are $X_1 = \{P_2\}$, $X_2 = \{P_1\}$, and $X_3 = \{P_1\}$. To ensure perfect correctness and fairness, P_1 should be allowed to send a message to P_2 because their invitation does not conflict with any other invitation. At the same time, both P_2 and P_3 should have the same opportunity to send a message to their desired recipient P_1 .

Generalizing this, an enhanced dialing protocol that ensures perfect correctness should accept all invitations for a recipient P_k whenever P_k has at most d invitations. If P_k receives more than d invitations, each invitation should have an equal chance of being accepted.

Below, we present our first protocol, Fusion, which is designed to minimize information leakage while adhering to the correctness and fairness rules outlined above.

4.1. Fusion: Protocol Description

Fusion⁴ builds on top of an anonymous posting protocol Π . There are known constructions for various settings, such as Atom [21], XRD [22], or Pi-butterfly [23].

Set up. During set up, each participant P_i sets up a public key pair (pk_i, sk_i) according to Π . Each party is identified by their public key. However, for readability, we will use P_i or BB (for the bulletin board) to refer to a party instead of their public key $(pk_i$ or $pk_{BB})$.

Algorithm for the participant. On input $X_i = \{X_{i,1}, X_{i,2}, \dots, X_{i,d}\}$, party P_i follows these steps:

- 1) **Generate invitations:** P_i generates d invitations for the invitees in their input. For each invitee $X_{i,j}$, P_i first generates a random number $r_{i,j}$ to serve as an identifier for the invitation and then forms the invitation $I_{i,j} = (r_{i,j}, X_{i,j})$.
- 2) **Send the invitations:** P_i participates in a run of the anonymous posting protocol Π to send the invitations, $I_{i,1}, I_{i,2}, \dots, I_{i,d}$, to the bulletin board. (P_i uses the public keys of the parties including the bulletin board to encrypt the invitations as specified by the anonymous posting protocol.)

4. The name “Fusion” reflects the way individual elements (invitations) come together to create a coordinated outcome, similar to a fusion-like reaction.

- 3) **Generate the output:** After the bulletin board publishes the results, P_i retrieves the parties associated with the identifiers, $r_{i,1}, r_{i,2}, \dots, r_{i,d}$, and outputs them as O_i .

Algorithm for the bulletin board. Let $I = \{I_{i,j} \mid i \in [N], j \in [d]\}$ be the multi-set of all invitation-invitee pairs.

After receiving all invitations via the anonymous posting protocol Π , the bulletin board follows these steps:

- 1) **Tally the invitations:** For each party P_i , the bulletin board generates the set $S_j \subseteq I$ containing the invitations for P_j . I.e., $S_j = \{r_{i,k} \mid \exists (r_{i,k}, P_j) \in I\}$.
- 2) **Assign invitations to invitees:** Let NO be the multi-set containing $h = d - |S_j|$ copies of P_j for each P_j with fewer than d invitations. Let an invitation be “accepted” if the bulletin board chooses it to be realized, and let NI be the set of invitations that are not accepted.
Both NO and NI are initially empty, and the bulletin board determines them as follows. For each set S_j :
 - If $|S_j| < d$ elements, the bulletin board accepts all invitations $(r_{i,k}, P_j) \in S_j$ and adds $d - |S_j|$ instances of P_j to NO .
 - If $|S_j| \geq d$ elements, the bulletin board randomly selects d of them to be accepted and moves the identifiers of the remaining invitations to NI .
- 3) **Assign remaining senders and invitees:** Next, the bulletin board randomly pairs each identifier $r_{i,j}$ in NI with an invitee P_k in NO . By construction, $|NI| = |NO|$, so this pairing is well-defined.
- 4) **Publish the results:** Let O be the set of accepted invitations, plus the randomly assigned identifier-recipient pairs. Finally, the bulletin board publishes O in random order. Figure 3 provides a visualization of our Fusion design.

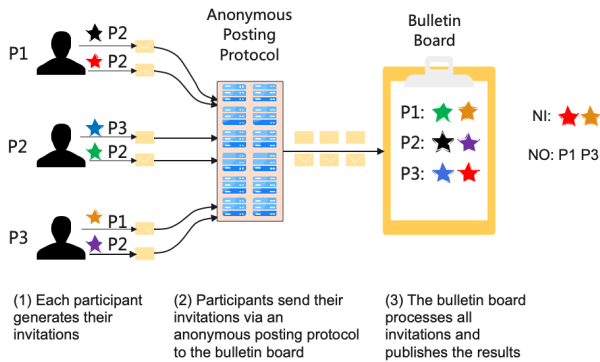


Figure 3: A visualization of Fusion (the differently colored stars represent independent randomness). *Note*, in this example, there are four invitations targeting P2; however, only two of them can be accepted. The remaining two (red star and brown star) is randomly assigned to either P1 or P3.

Note: Some other dialing protocols [19], [25], [26], [28], [30], [31] include key-exchange functionality. We note that adding this feature in Fusion is straightforward: simply encrypt the key (or any shared secret required for the sub-

sequent communication protocol) under the invitee’s public key (rather than a random identifier), place that ciphertext in the invitation, and let the bulletin board publish it.

4.2. Fusion: Analysis

We first argue that Fusion achieves perfect correctness and fairness, and then characterize precisely what information is leaked to an active adversary. We show that this leakage is limited to the input pattern of corrupted parties and the popularity pattern of honest parties, and that this leakage is unavoidable for any perfectly correct and fair protocol.

Clearly, Fusion protocol achieves perfect correctness and fairness. If at most d participants select the same invitee, the protocol accepts all of these invitations. However, if more than d participants select the same invitee, the protocol randomly selects and accepts d of these invitations. This method maximizes the number of accepted invitations, ensuring perfect correctness. Fairness is maintained because, when more than d participants choose the same invitee, each participant has an equal chance of being selected.

In the remainder of this section, we will provide a detailed security analysis of the Fusion protocol, starting with the following observations.

Hiding the senders’ identities during the invitation phase: At the start of each round, each sender generates a random identifier for each invitation, replacing their actual identity. Neither other participants nor the bulletin board can associate $r_{i,j}$ with P_i . During processing, the bulletin board treats $r_{i,j}$ as the sender’s identifier but cannot trace it back to P_i .

Unlinking multiple invitations from a single participant: If $d > 1$, meaning each participant submits multiple invitations, P_i generates d distinct identifiers. This ensures that neither the bulletin board nor other participants can link these invitations to the same sender. To an external observer, including the bulletin board, each invitation appears unlinkable to any other invitation.

Unlinking participants from invitations: To further obscure the connection between participants and their invitations, each invitation is submitted via an anonymous posting protocol. As a result, when the bulletin board receives invitations, it cannot determine their senders. Additionally, because invitations are encrypted using the bulletin board’s public key pk_{BB} , their contents remain secure in transit.

Concealing the senders’ identities during the announcement phase: When the bulletin board publishes the results, the invitations are not encrypted, but only the owner recognizes their identifier. Consequently, the sender’s identity remains hidden even from the invitee.

4.2.1. Leakage Functions

Although Fusion prevents an adversary from linking invitations to their senders or other invitations from the same source, some information is still revealed. Below, we define

specific leakage functions for qualifying the security level of the protocol.

Leakage observed by the corrupted bulletin board. A corrupted bulletin board does not learn the senders of invitations but can observe the “popularity” of each recipient. Specifically, it obtains the *popularity pattern*.

Let Bad be the set of corrupted parties, and let CI be the set of their indices. For example, if $\text{Bad} = \{P_1, P_3, P_8\}$, then $CI = \{1, 3, 8\}$.

$$\text{PopularityPattern}(\vec{X}) = \{(P_j, C_j^h)\}_{j \in [N]},$$

where C_j^h represents the number of times recipient P_j appears in the *honest* parties’ inputs, i.e., $C_j^h = \sum_{i \in [N] \setminus CI} C_{i \rightarrow j}$, where $C_{i \rightarrow j}$ denotes the number of times P_j appears in X_i (as established in Section 3.2).

Leakage observed by corrupted parties. Now, consider the perspective of a corrupted participant P_i . Since P_i has access to both its input X_i and output O_i , leakage arises from these inputs and outputs.

We define the *input pattern* as:

$$\text{InputPattern}(X_1, X_2, \dots, X_N) = \{(P_i, C_i)\}_{i \in CI}.$$

This simply records the inputs of corrupted parties. However, corrupted parties may gain additional insights beyond their direct outputs.

Let us revisit our earlier example with three participants ($N = 3$) and one invitation per party ($d = 1$). The invitations are as follows: $X_1 = \{P_2\}$, $X_2 = \{P_1\}$, and $X_3 = \{P_1\}$.

First, let us assume that P_3 is corrupted. Fusion can assign P_2 ’s invitation to P_3 and P_3 ’s invitation to P_1 . Alternatively, it can assign P_2 ’s invitation to P_1 and P_3 ’s invitation to P_1 . In the latter case, P_3 can observe that an honest party also wanted to communicate with P_1 , and no one wanted to talk with P_3 , providing a “window” into the inputs of the honest parties.

Now, consider an alternative setting where P_1 is corrupted and P_3 is not. There is no “window” into the honest participants’ inputs, as P_1 ’s input does not conflict with those of the other participants.

From the above insight, we define the following. Let Index be a function that takes a party and returns the index of the party’s identifier, i.e., $\text{Index}(P_i) = i$. Let:

$$WS_{i,j} = \begin{cases} C_{i \rightarrow \text{Index}(X_{i,j})} & \text{if } C_{\text{Index}(X_{i,j})} > d \\ 0 & \text{otherwise} \end{cases}.$$

$$WS_i = (WS_{i,1}, \dots, WS_{i,d})$$

$$\text{WindowSizes}(\vec{X}) = \{(P_i, WS_i)\}_{i \in CI}$$

If all values in $\text{WindowSizes}(\vec{X})$ are zero or the set is empty, we define the *window pattern* as $\text{WindowPattern}(\vec{X}) = \perp$. Otherwise:

$$\text{WindowPattern}(\vec{X}) = (\text{WindowSizes}(\vec{X}), NO),$$

where NO is a multi-set containing $h = d - C_j$ copies of each P_j that received at most d invitations. Intuitively,

window leakage arises when a corrupted party’s invitation competes with honest invitations for an oversubscribed recipient.

4.2.2. Security statement

Here, we prove that Fusion leaks only the input and the popularity patterns when the bulletin board and some of the participants are corrupted.

Theorem 2. *Fusion is secure w.r.t. the leakage function, $L(\vec{X}) = (\text{InputPattern}(\vec{X}), \text{PopularityPattern}(\vec{X}))$, against the collusion of the bulletin board and corrupted participants.*

Proof idea. The proof proceeds by reduction from the dialing game to the anonymity game. We assume there exists an adversary $\mathcal{A}$ that can win the dialing game against Fusion w.r.t. the leakage function with non-negligible advantage. Then, we construct a reduction $\mathcal{B}$ that uses $\mathcal{A}$ as a subroutine to break the anonymity game. Intuitively, $\mathcal{B}$ acts as a challenger in the dialing game while simultaneously playing the anonymity game with its own challenger. It converts the inputs X^0 and X^1 in the dialing game into the inputs of the anonymity game $\vec{\sigma}^0$ and $\vec{\sigma}^1$ so that all messages of all honest participants’ are generated consistently with the Fusion rules. Since the leakage function only includes the input and popularity patterns, the two input vectors have identical popularity and input distributions, so the view of the bulletin board and all corrupted participants remains indistinguishable. Consequently, $\mathcal{B}$ wins the anonymity game whenever $\mathcal{A}$ succeeds in distinguishing between X^0 and X^1 , contradicting the assumed anonymity of Fusion. See Appendix A for the full proof. $\square$

Remark 1. *When combined with a fully secure conversation protocol such as Pi-butterfly [23], which leaks only the input pattern, Theorem 2 implies that sender anonymity can be achieved while leaking at most the input pattern and the popularity pattern. This guarantee holds in a synchronous, static network with an active adversary corrupting a constant fraction $< \frac{1}{2}$ of the parties (possibly including the bulletin board).*

Remark 2. *At first glance, it may seem surprising that the leakage function in Theorem 2 does not include the window pattern. However, this becomes clearer when we consider that the window pattern can be inferred from the popularity pattern.*

4.3. Matching Lower Bound

As explained in the previous section, Fusion is perfectly correct, fair, and leaks at most the popularity pattern (and the input pattern). Here, we prove a matching lower bound: we show that any enhanced dialing protocol that is perfectly correct and fair (whether it uses a bulletin board or an entirely different design), may “leak the popularity pattern.” More specifically, we show that:

Theorem 3. Let $L : \mathcal{X} \rightarrow \mathcal{L}$ be a leakage function such that there exist two input vectors $\vec{X}, \vec{X}' \in \mathcal{X}$ satisfying $\vec{X} \equiv_{\text{Bad}} \vec{X}'$, $L(\vec{X}) = L(\vec{X}')$, and $\text{PopularityPattern}(\vec{X}) \neq \text{PopularityPattern}(\vec{X}')$. Then there exists no enhanced dialing protocol that is secure with respect to L , perfectly correct, and fair, even against a passive adversary corrupting a single participant.

Proof. Let P_N be the corrupted party. Let $\vec{X}^0$ and $\vec{X}^1$ be neighboring input vectors with the following properties:

- In both $\vec{X}^0$ and $\vec{X}^1$: all remaining participants, $P_1, \dots, P_{N-1}$, are honest and send invitations exclusively to themselves. Consequently, P_N does not receive any invitations from honest parties.
- In $\vec{X}^0$: every honest party receives exactly d invitations. Thus, when we disregard the corrupted party P_N , the popularity pattern is given by $C_1 = C_2 = \dots C_{N-1} = d$ where C_i denotes the number of times that P_i shows up in the honest parties' inputs.
- In $\vec{X}^1$: the distribution is slightly modified: one party receives $d-1$ invitations while another receives $d+1$ invitations. For concreteness, let us assume that P_1 receives $d-1$ invitations and P_2 receives $d+1$ invitations.
- In both $\vec{X}^0$ and $\vec{X}^1$: the input for P_N is set to $\underbrace{\{P_2, P_2, \dots, P_2\}}_{d \text{ times}}$.

Below, we show that the adversary can distinguish between the scenario given by $\vec{X}^0$ (Scenario 0) and the alternative one given by $\vec{X}^1$ (Scenario 1) with a non-negligible advantage.

Under Scenario 0, since P_1 receives d invitations from honest parties, a perfectly correct enhanced dialing protocol would ensure that P_1 never appears in the output observed by the adversary $\mathcal{A}$. In contrast, in Scenario 1, because P_1 receives only $d-1$ invitations, there is a non-zero probability that $\mathcal{A}$ will observe P_1 in its output.

Thus, the adversary's advantage can be expressed as:

$$\begin{aligned} \text{Adv} &= \left| \Pr[\mathcal{A} \text{ outputs } 1 \mid \vec{X}^1] - \Pr[\mathcal{A} \text{ outputs } 1 \mid \vec{X}^0] \right| \\ &= \Pr[\mathcal{A} \text{ sees } P_1 \text{ in the output} \mid \vec{X}^1]. \end{aligned}$$

Now, consider the random assignment process. The probability that a single invitation is randomly assigned (i.e., not given to the original invitee) is $\frac{d+1}{2d+1}$. Given that an invitation is randomly assigned, the probability that it is assigned to P_1 is $\frac{1}{d+1}$.

Since there are d invitations from P_N , the probability that $\mathcal{A}$ observes P_1 in the output under Scenario 1 is

$$d \cdot \frac{d+1}{2d+1} \cdot \frac{1}{d+1} = \frac{d}{2d+1}.$$

From above, the adversary's advantage is non-negligible for fixed d and approaches one-half for large values of d . $\square$

In the proof above, the input vectors $\vec{X}^0$ and $\vec{X}^1$ are neighboring. Thus, we also showed that:

Corollary 1. No perfectly correct and fair enhanced dialing protocol can achieve differential privacy.

Note: that these lower bounds hold even against a significantly weaker adversary, strengthening the impossibility result.

In Theorem 3, we demonstrated that given a perfectly correct and fair enhanced dialing protocol Σ , there are scenarios where Σ cannot prevent an adversary from gaining more information about the popularity pattern. We showed that if the adversary knows in advance that the input vector is either $\vec{X}^0$ or $\vec{X}^1$ (as defined in the proof of Theorem 3), then they can distinguish between these two situations with non-negligible advantage.

By generalizing the proof, we can strengthen this impossibility result to show that the adversary retains a non-negligible advantage even when the inputs $\vec{X}^0$ and $\vec{X}^1$ are selected randomly.

As long as two inputs have different popularity patterns, the adversary can distinguish them. If $\text{PopularityPattern}(\vec{X}^0) \neq \text{PopularityPattern}(\vec{X}^1)$, there is some party P_i who receives more honest invitations under $\vec{X}^0$ than under $\vec{X}^1$. The adversary then has all corrupted parties send their invitations exclusively to P_i . Because P_i receives fewer honest invitations in $\vec{X}^1$, the probability that there are d invitations from corrupted parties are realized in the output is strictly higher under $\vec{X}^1$ than under $\vec{X}^0$. This gap is non-negligible.

4.3.1. Recovering the Popularity Pattern

Here, we demonstrate that it is possible to efficiently recover the popularity pattern even without prior knowledge of the input.

Our algorithm. To do this, we first developed an efficient algorithm, FindPopularityPattern, to recover the popularity pattern.

We assume a setting in which the input vector $\vec{X}$ remains fixed across R runs of the enhanced dialing protocol Σ .

FindPopularityPattern is given the inputs and outputs of the corrupted parties corresponding to $\vec{X}$ and $\vec{O}_1, \vec{O}_2, \dots, \vec{O}_R \leftarrow \Sigma$; and from these, it produces an estimate of the popularity pattern, denoted $\hat{C}_1^h, \hat{C}_2^h, \dots, \hat{C}_N^h$.

To clarify the algorithm's structure, we define the following subsets of parties. Let U_1 be the parties that appear in the corrupted inputs and have more than d invitees (i.e., $C_i > d$), let U_2 be the parties that appear in the corrupted inputs and have exactly d invitees (i.e., $C_i = d$), let U_3 be the parties that have fewer than d invitees (i.e., $C_i < d$), and let U_4 be the parties that have at least d invitees but do not appear in the corrupted inputs. The full set of parties is thus $\mathcal{P} = U_1 \cup U_2 \cup U_3 \cup U_4$.

Let $\vec{A} = (A_1, \dots, A_n)$, where A_i denotes, for corrupted parties, the total number of invitations involving party P_i realized over R rounds. Let $\vec{k} = (k_1, \dots, k_n)$, where k_i denotes the number of invitations that corrupted parties send to party P_i . Here, R denotes the total number of rounds, and d denotes the maximum number of invitations each party can send per round.

The algorithm consists of three subroutines PPatternU1, PPatternU23, and PPatternU4.

For each run, we first determine U_1 by observing which parties appear fewer times in the outputs compared with the inputs and then estimate C_i for each party P_i in U_1 from this difference (the frequency that P_i appears in the inputs minus the frequency that P_i appears in outputs); for details, see **Algorithm 1** PPatternU1.

Algorithm 1 PPatternU1

Require: $\vec{A}, \vec{k}, d, R$

Ensure: U_1 and estimated values of C_i for parties in U_1

- 1: Initialize $U_1 \leftarrow \emptyset$
 - 2: **for** $i = 1$ to n **do**
 - 3: **if** $i \notin CI$ and $k_i > 0$ and $A_i < k_i \cdot R$ **then**
 - 4: Append i to U_1
 - 5: Estimate $C_i \leftarrow \text{Round}\left(\frac{k_i \cdot R \cdot d}{A_i}\right)$
 - 6: **end if**
 - 7: **end for**
-

Similarly, we directly estimate C_i for each party P_i in U_3 by comparing the frequencies that P_i appear in the inputs with their frequencies in the outputs. U_2 is then determined from U_1 and U_3 ; see **Algorithm 2** PPatternU23.

Algorithm 2 PPatternU23

Require: $\vec{B}, \vec{k}, d, U_1$

Ensure: U_2, U_3, U_4 and proportional estimates of C_i for parties in U_3 and exact values for U_2

- 1: Initialize $U_2, U_3, U_4 \leftarrow \emptyset$
- 2: **for** $i = 1$ to n **do**
- 3: **if** $i \notin CI$ and $B_i > 0$ **then**
- 4: Append P_i to U_3
- 5: **end if**
- 6: **end for**
- 7: **for** $i = 1$ to n **do**
- 8: **if** $i \notin CI$ and $k_i > 0$ and $P_i \notin U_1 \cup U_3$ **then**
- 9: Append P_i to U_2
- 10: Set $C_i \leftarrow d$
- 11: **end if**
- 12: **end for**
- 13: **for** $i = 1$ to n **do**
- 14: **if** $i \notin CI$ and $P_i \notin U_1 \cup U_2 \cup U_3$ **then**
- 15: Append P_i to U_4
- 16: **end if**
- 17: **end for**
- 18: Let $U_3 = \{P_a, P_b, P_c, \dots\}$
- 19: **Proportional relation:**

$$B_a : B_b : B_c : \dots \approx (d - C_a) : (d - C_b) : (d - C_c) : \dots$$

If $U_1 \cup U_2 \cup U_3 = \mathcal{P}$ (i.e., $U_4 = \emptyset$) and the sample size is sufficiently large (provided by increasing R), it is possible to accurately estimate all C_i -values using Algorithms PPatternU1 and PPatternU23.

Conversely, if $U_4 \neq \emptyset$, it is not possible to directly estimate C_i for the parties in U_4 . **Algorithm 3** PPatternU4

approximates these values based on the bounds and proportional constraints inferred from U_3 .

Algorithm 3 PPatternU4

Require: U_3, U_4 , and the three constraints

Ensure: Estimated values of C_i for parties in U_3 and U_4

- 1: Use integer linear programming to compute the maximum feasible C_i values for U_3 ,
subject to observed replacement behavior and total count bounds.
 - 2: Compute $\text{Sum}_{U_4} \leftarrow \text{Sum}_{U_3+U_4} - \text{Sum}_{U_3}$
 - 3: **for** each $P_i \in U_4$ **do**
 - 4: Set $C_i \leftarrow \text{Round}\left(\frac{\text{Sum}_{U_4}}{|U_4|}\right)$
 - 5: **end for**
-

Evaluating our algorithm. We make no assumption on the internal mechanics of the enhanced dialing protocol Σ , assuming only that Σ is both perfectly correct and fair. Under these assumptions, each input vector induces a fixed distribution over outputs.

To generate test instances, we randomly selected an input vector and then sampled corresponding output vectors according to the protocol's fairness constraints. Specifically, if a participant received no more than d invitations, all were included in the output. If a participant received more than d invitations, only d were accepted, each selected uniformly at random. The remaining invitations were redistributed among undersubscribed participants (that is, those with fewer than d invitations), also uniformly at random.

Each test ran FindPopularityPattern over R rounds, where R ranged from 1 to 100. The algorithm was provided with the corrupted parties' inputs (fixed for R rounds) and outputs for each round generated using the procedure described above.

We evaluated performance across several parameters. The total number of parties, N , was set to either 100 or 50,000; the fraction of corrupted parties, $\kappa = 1 - h$, was chosen between 0.03 and 0.10; and the number of invitations per party, d , was chosen from the set $\{1, 2, 5, 10, 20\}$.

For each test, we computed the error rate as:

$$\text{error rate} = \frac{\|\text{PopularityPattern}(\vec{X}) - (\hat{C}_1^h, \hat{C}_2^h, \dots, \hat{C}_N^h)\|_1}{\|\text{PopularityPattern}(\vec{X})\|_1}$$

where $\hat{C}_1^h, \hat{C}_2^h, \dots, \hat{C}_N^h \leftarrow \text{FindPopularityPattern}$ denotes our estimates. We computed accuracy as: accuracy = 1 - error rate.

We found that it is possible to recover the popularity pattern even with "small samples." For instance, accuracy is at least 0.8 for $N \geq 100$ participants, corruption rate $\kappa \geq 0.03$, and $R \geq 20$. See Figure 4.

We observe in Figures 4(b)-(c) that, despite the dramatic difference in the number of participants (one plot uses $N = 100$ and the other uses $N = 50,000$), the curves are nearly identical. This occurs because, in both cases, the corruption rate, κ , is the same. So, the adversary sees the same relative proportion of information regardless of

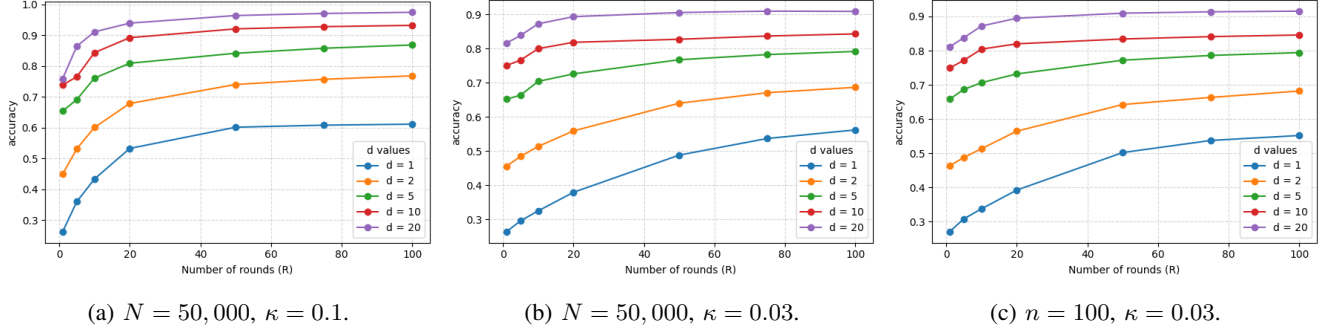


Figure 4: Accuracy of our algorithm, FindPopularityPattern, for various settings specified by the number of participants, N ; the fraction of corrupted parties, κ ; the number of invitations per party, d ; and the number of runs, R . Accuracy increases as N , κ , d , or R increases.

N . Consequently, the adversary’s uncertainty decreases at a similar rate as the rounds progress. These experiments demonstrate that popularity leakage is not merely theoretical, but efficiently exploitable in practice.

5. Fusion+

Previously, we introduced Fusion, a simple enhanced dialing protocol that is perfectly correct and fair. We demonstrated that Fusion is secure with respect to the input and popularity patterns (Theorem 2). We established that these patterns necessarily leak for any protocol that is both perfectly correct and fair (Theorem 3). Moreover, it is not possible to achieve even differential privacy if we aim for perfect correctness and fairness (Corollary 1).

In this section, we explore a modification to Fusion that trades off some correctness in order to obscure the popularity pattern. We refer to this modified protocol as Fusion+. Instead of each participant P_i sending their input $X_{i,1}, \dots, X_{i,d}$ directly to the bulletin board (as separate messages), they first process their invitations through the Randomized Response mechanism [42] and then send the outputs of this mechanism, denoted as $Y_{i,1}, \dots, Y_{i,d}$, to the bulletin board. This ensures that the number of invitations received by each participant becomes noisy, preventing the bulletin board from learning the true popularity pattern. By incorporating a differentially private mechanism into the process, we ensure that the observed popularity pattern becomes noisy.

This effectively leverages the fact that participants use an *anonymous channel* to send their noisy inputs to the bulletin board. As a result, we find ourselves in the shuffle model, which requires significantly less noise than applying Randomized Response directly in the local model.

5.1. Fusion+: Protocol Description

Here, we detail the differences between Fusion and Fusion+. In Fusion+, we introduce an additional parameter, p , for the Randomized Response mechanism; this is the

probability of replacing an invitee $X_{i,j}$ with a random recipient.⁵

Set up. Like in Fusion, each participant P_i generates a public key pair (pk_i, sk_i) and makes the public portion available via a PKI.

Algorithm for the participant. The algorithm for P_i includes the additional step (2), in which P_i runs each $X_{i,j}$ through the Randomized Response mechanism. After this, Fusion+ proceeds as Fusion.

Specifically, on input $X_i = \{X_{i,1}, X_{i,2}, \dots, X_{i,d}\}$, party P_i follows these steps:

- 1) **Generate invitations:** Step 1 in Fusion.
- 2) **Add noise to the invitation:** For each invitation $I_{i,j}$, sample $Y_{i,j}$ randomly; $Y_{i,j}$ is set to $X_{i,j}$ with probability $1 - p$, and with probability p , it is replaced by a uniformly random participant $X'_{i,j} \leftarrow \mathcal{P}$. This results in a modified (noisy) invitation: $I'_{i,j} = (r_{i,j}, Y_{i,j})$.
- 3-4) **Send the invitations and generate the output:** Steps 2-3 in Fusion.

Algorithm for the bulletin board. The bulletin board follows the same procedure as that in Fusion.

5.2. Fusion+: Analysis

Fusion+ ensures both strong security and correctness while maintaining fairness. We also demonstrate that the differentially private mechanism in Fusion acts like a “tunable knob,” enabling a smooth and measurable trade-off between the privacy budget and correctness.

5.2.1. Fusion+: Fairness

First, to understand why Fusion+ is fair, consider that while Fusion+ converts some actual invitations into “noisy” ones, this transformation does not compromise fairness.

The probability that any invitation $X_{i,j}$ is replaced by noise is independent and occurs with a fixed probability p . Like in Fusion, Fusion+ accepts all (noisy) invitations for a recipient P_j if the total number C_j of (noisy) invitations for P_j does not exceed d . Otherwise, a random sample of size

5. The Randomized Response mechanism effectively adds a random number of dummy invitations for every recipient, sampled according to a binomial distribution.

d is selected from these (noisy) invitations. This approach ensures that every invitation for the same invitee has the same chance of being realized. We formalize this intuition in Appendix B.

5.2.2. Fusion+: Differential Anonymity

Next, we present our differential privacy result.

There are N parties, of which hN are honest. Every party issues d invitations. Thus, there are hNd honest participants who collectively issue hNd invitations. Each invitation is replaced with noise with fixed probability p .

Theorem 4. *For all $0 < h \leq 1$, $\epsilon \leq 1$, and $0 < \delta \leq 1$, Fusion+ satisfies (ϵ, δ) -differential privacy provided that:*

$$p \geq \max \left\{ \frac{14N \log(2/\delta)}{(hNd - 1)\epsilon^2}, \frac{27N}{(hNd - 1)\epsilon} \right\}.$$

Proof idea. Previously, we showed that Fusion is secure with respect to the input pattern and the popularity pattern (Theorem 2). In other words, since the input pattern leakage is inherent to the problem, the leakage is at most the popularity pattern. Therefore, to show that Fusion+ is differentially private, it suffices to show that the Randomized Response mechanism effectively conceals the popularity pattern.

We can reframe the popularity pattern as a histogram $C_1^h, C_2^h, \dots, C_N^h$ of invitations from the honest parties, where each C_j^h indicates the number of times party P_j appears in the inputs of the honest parties. Because the noisy invitations are sent to the bulletin board anonymously, we can apply Theorem 1 (in Section 2.3), a known result for differentially private histograms in the shuffle model, to bound the noise required to achieve the desired level of privacy. See Appendix C for the full proof. $\square$

In the proof of Theorem 4, a more exact analysis allows us to tighten the above bound and obtain the following corollary.

Corollary 2. *For all $0 < h \leq 1$, and $0 < \delta \leq 1$, Fusion+ satisfies (ϵ, δ) -differential privacy provided that $p \geq \max \left\{ \frac{3N \log(2/\delta)}{(hNd - 1)(e^{\epsilon/2} - 1 - \frac{N}{p(hNd - 1)})^2}, \frac{2N \log(2/\delta)}{(hNd - 1)(1 - e^{-\epsilon/2})^2} \right\}$.*

Remark 3. *When combined with Pi-butterfly [23], which is fully secure against an active adversary corrupting a constant fraction $< \frac{1}{2}$ of the parties in a static, synchronous network, this result implies that sender anonymity can be achieved in a differentially private manner. In particular, beyond the leakage inherent to the conversation protocol, the dialing phase reveals no additional information except what is permitted by the chosen differential privacy parameters.*

5.2.3. Fusion+: Correctness

Finally, we show:

Theorem 5. *Fusion+ is (α, β) -correct when*

$$p \leq \alpha - \frac{\log\left(\frac{1}{\beta}\right)}{Nd} - \frac{\sqrt{\log\left(\frac{1}{\beta}\right) \left(2Nd(1 - \alpha) - \log\left(\frac{1}{\beta}\right)\right)}}{Nd}.$$

Proof. Recall that C_j is the number of times P_j appears in the input vector $\vec{X}$, and M_j is the total number of invitations for P_j that are fulfilled in the outputs.

Let Q_j be the number of invitations for P_j that are *not* replaced with noise, and let $Q = \sum_{j=1}^N Q_j$ be the total number of invitation not replaced with noise. We want to bound (above by β) the following expression: $\Pr\left[\frac{\sum_{j=1}^N M_j}{\sum_{j=1}^N \min(C_j, d)} < 1 - \alpha\right]$. Since $Q_j \leq M_j$ and $\min(C_j, d) \leq C_j$, we can do so by bounding, instead, the expression $\Pr\left[\frac{\sum_{j=1}^N Q_j}{\sum_{j=1}^N C_j} < 1 - \alpha\right] = \Pr[Q < (1 - \alpha)Nd]$.

Q can be modeled by a binomial random variable with parameters Nd and $1 - p$, i.e., $Q \leftarrow \text{Bin}(Nd, 1 - p)$. Thus, from a multiplicative Chernoff bound [43, Theorem 4.5], when $p \leq \alpha$, it follows that

$$\begin{aligned} \Pr[Q < (1 - \alpha)Nd] &= \Pr\left[Q < \left(1 - \frac{\alpha - p}{1 - p}\right) \mathbb{E}[Q]\right] \\ &\leq \exp\left(-\frac{(\alpha - p)^2 Nd}{2(1 - p)}\right) \end{aligned}$$

To guarantee (α, β) -correctness it suffices to choose p such that $\exp\left(-\frac{(\alpha - p)^2 Nd}{2(1 - p)}\right) \leq \beta$; in other words, by satisfying the constraint: $p^2 + \left(\frac{2 \ln(1/\beta)}{Nd} - 2\alpha\right)p + \left(\alpha^2 - \frac{2 \ln(1/\beta)}{Nd}\right) \geq 0$.

For $p \leq \alpha$, this corresponds to the region specified in Theorem 5. $\square$

5.2.4. Fusion+: Numerical Results

We first analyze how much privacy a given level of noise can provide in Fusion+. We then evaluate the trade-off between privacy and correctness.

Achievable ϵ values. We first examine how the probability p affects the privacy budget ϵ in various settings for the number of participants (N), the lower bound on the fraction of honest parties (h), and the number of invitations per party per run (d). To do this, we use our bound from Corollary 2 instead of Theorem 4 because it provides a tighter bound and handles the case $\epsilon > 1$.

We fix $\delta = 0.0001$, and we consider the parameter settings that allow for a privacy budget ϵ within the range of 0.01 to $\ln 3$, which corresponds to values considered reasonable in practice [44]. (To emphasize meaningful security guarantees, we limit the vertical axis of our plots to $\epsilon = \ln 5$.) Fixing all other parameters (h , d , and p), we find that varying N has little impact on ϵ . For example, when $hd = 300$, $p = 0.5$, and $\delta = 10^{-4}$, the computed values of ϵ are 0.90336 for $N = 10$, and 0.90317 for $N = 50,000$, resulting in a difference of only 0.00019. This shows that ϵ is effectively independent of N and is determined primarily by the product hd , which is in line with our analytical derivations. See Figure 5 that plot e^ϵ as a function of p .

To achieve the reasonable privacy level of $\epsilon = \ln 3$, for $hd = 150$, we require a noise probability of at least $p = 0.739$. Fixing $h = 0.9$ then implies that each participant

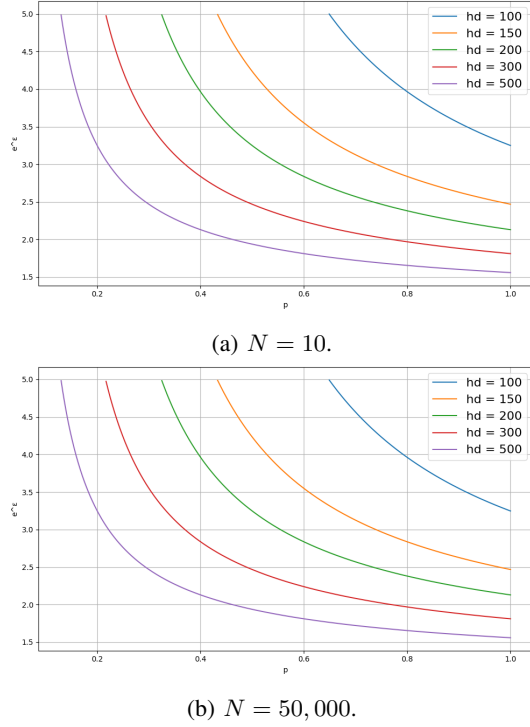


Figure 5: Privacy budget e^ϵ plotted against probability p .

must send roughly $d = 166$ invitations per round. Given this noise probability, each honest party sends, on average, 43 real invitations and 123 noise invitations to the bulletin board per run. This level of communication is reasonable in practice, as modern devices commonly interact with dozens to hundreds of IP addresses or domains per second.

Next, let us consider the case where a whistleblower wants to make an anonymous report, which requires sending only one real invitation, i.e., $(1 - p) \cdot d = 1$. In such scenarios, achieving strong privacy (e.g., $\epsilon = \ln 3$) requires a large hd value. This means that even to send a single real invitation while ensuring a meaningful privacy guarantee, the whistleblower must also generate a large number of noise invitations. This cost is justified by the importance of preserving anonymity. For example, when $N = 50000$, $h = 0.9$, and $\delta = 0.0001$, satisfying both $(1 - p) \cdot d = 1$ and $\epsilon \leq \ln 3$ requires setting $d = 124$ and $p = 0.9919$.

More generally, for a fixed value of hd , increasing p introduces more noise, which results in fewer real invitations. On the other hand, lowering ϵ (and δ) increases privacy. Conversely, when p is held constant, increasing hd (meaning sending more total invitations per round) also reduces ϵ . Therefore, to maintain a bounded ratio of noise to real invitations while still achieving a strong privacy guarantee, one must either increase the fraction of honest parties (h) or allow for a greater number of invitations per round (d).

Correctness-anonymity trade-off. Achieving meaningful privacy requires $hd \geq 100$; see Figure 5.

Now, let us examine the correctness bound given in Theorem 5 for $\beta = 0.05$ and $Nd = 100$. The curve lies

almost exactly on the line $\alpha = p$, since the correction term $\frac{\log(\frac{1}{\beta})}{Nd} + \frac{\sqrt{\log(\frac{1}{\beta})(2Nd(1-\alpha) - \log(\frac{1}{\beta}))}}{Nd}$ is very small. In a realistic deployment, with at least 10 parties, Nd easily exceeds 100 (and could reach 10,000 when targeting $\epsilon = 1$). That makes the correction term even closer to zero, so $\alpha \approx p$. In other words, the correctness parameter α is essentially a linear, positively correlated function of p : as honest parties inject more noise (higher p), the enhanced dialing protocol’s correctness parameter α increases, leading to weaker correctness.

Plugging in the parameters from our two examples yields compact and interpretable bounds that illustrate different application scenarios. In Case 1, which models a typical communication setting, we set $hd = 150$, $h = 0.9$, $p = 0.739$, $N = 50,000$, and $\delta = 10^{-4}$. Under these parameters, the correctness guarantee is $\alpha = 0.73943$, $\beta = 0.05$, and the privacy guarantee is $\epsilon = \ln 3$, $\delta = 10^{-4}$. In Case 2, reflecting a more sensitive context such as anonymous whistleblowing, we choose $N = 50,000$, $h = 0.9$, $d = 124$, $p = 0.9919$, and $\delta = 10^{-4}$. Under these parameters, the correctness guarantee becomes $\alpha = 0.991988$, $\beta = 0.05$, and the privacy guarantee is $\epsilon = \ln 3$, $\delta = 10^{-4}$.

This further highlights a clear trade-off, achieving stronger privacy (i.e., a higher p) comes at the cost of reduced correctness, as α approaches 1, meaning that the proportion of real invitations becomes very low. In contrast, a more moderate noise level (e.g., $p = 0.739$) can still ensure $\epsilon = \ln 3$ while maintaining relatively better correctness (e.g., $\alpha \approx 0.739$, $\beta = 0.05$).

Conclusion. We introduce Fusion and Fusion+, two protocols designed to ensure sender anonymity while overcoming the conventional input/output restrictions present in existing dialing protocols. Fusion minimizes information leakage to only what is unavoidable in any perfectly correct and fair solution. Fusion+ builds upon Fusion by integrating a differential privacy mechanism, which helps to obscure this leakage. It demonstrates strong privacy guarantees in typical network environments.

Although we explain our ideas simply (using a single server with a bulletin board), we remark that our general design could be modified to include multiple servers (to improve practical considerations such as fault tolerance), and this change could only improve the security guarantees.

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Appendix A.

Fusion: Proof of Indistinguishability

Proof of Theorem 2. We assume, for the sake of contradiction, that our enhanced dialing protocol is not secure (Definition 2) w.r.t. the leakage function with InputPattern and PopularityPattern.

This assumption implies the existence of a probabilistic polynomial-time (p.p.t.) adversary $\mathcal{A}$ capable of winning the dialing game with a non-negligible probability. That is, there are input vectors $\vec{X}^0, \vec{X}^1 \equiv_{\text{Bad}, (\text{InputPattern}, \text{PopularityPattern})} \vec{X}^0$, such that $\mathcal{A}$ can distinguish whether they are interacting in a run on input $\vec{X}^0$ or input $\vec{X}^1$ with non-negligible advantage.

We construct the reduction $\mathcal{B}$ as follows. (*Note:* $\mathcal{B}$ serves as the challenger in the dialing game against $\mathcal{A}$ while simultaneously acting as the adversary in the anonymity game against their challenger $\mathcal{C}$.)

- 1) $\mathcal{A}$ “chooses” the set $\text{Bad} \subseteq \{P_1, P_2, \dots, P_N, \text{BB}\}$ of parties to corrupt and sends Bad to $\mathcal{B}$. $\mathcal{B}$ forwards this to $\mathcal{C}$.
- 2) $\mathcal{C}$ generates the keys for the honest parties and sends the public portions of the keys, $\text{pk}(\text{Honest})$, to $\mathcal{B}$. $\mathcal{B}$ forward them to $\mathcal{A}$.
- 3) $\mathcal{A}$ sends the public keys of the corrupted parties, $\text{pk}(\text{Bad})$, to $\mathcal{B}$, who then forwards them to $\mathcal{C}$.
- 4) $\mathcal{A}$ (the adversary in the dialing game) selects $\vec{X}^0, \vec{X}^1$ (equivalent w.r.t. the two leakage functions) and sends

them to $\mathcal{B}$ (the challenger in the dialing game). See the dialing game in Section 3.1.

- 5) $\mathcal{B}$ converts $\vec{X}^0$ and $\vec{X}^1$ into the challenge inputs for the anonymity game, $\vec{\sigma}^0$ and $\vec{\sigma}^1$ as follows:

- a) For each P_i , let C_i^h be the number of occurrences of P_i in the honest parties’ inputs. (This is the same number of occurrences of P_i in $\vec{X}^1$.) For each recipient P_i , $\mathcal{B}$ samples $r_{i,1}, r_{i,2}, \dots, r_{i,C_i^h}$ randomly.
- b) Let $\text{Position}(\vec{X}, i, j)$ be a function that takes an input vector $\vec{X}$ and an invitee $X_{i,j} = P_k$ and outputs ℓ where $X_{i,j}$ is the ℓ^{th} occurrence of P_k in the honest parties’ inputs. For all $b \in \{0, 1\}$, $i \in [N] \setminus CI$, $j \in [d]$, $\mathcal{B}$ forms the invitations, $L_{i,j}^b = (r_{i, \text{Position}(\vec{X}^b, i, j)}, X_{i,j})$. For $b \in \{0, 1\}$, $\mathcal{B}$ sets $\vec{\sigma}^b = (\sigma_1^b, \sigma_2^b, \dots, \sigma_N^b, \sigma_{\text{BB}}^b)$, where $\sigma_{\text{BB}}^b = \{\}$ and, for all $i \in [N]$,

$$\sigma_i^b = \begin{cases} \{\}, & \forall i \in CI \\ \{L_{i,1}^b, L_{i,2}^b, \dots, L_{i,d}^b\}, & \forall i \in [N] \setminus CI. \end{cases}$$

See the Section 2.2 for a description of the anonymity game.

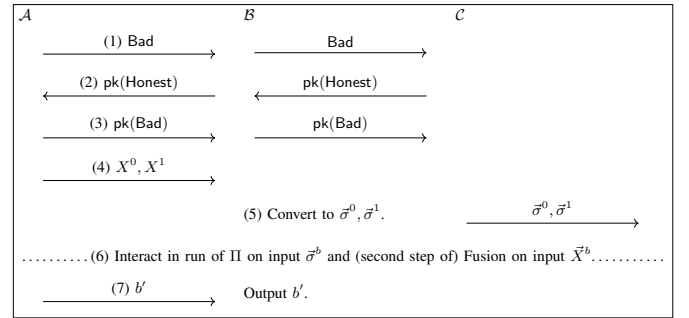


Figure 6: Structure of reduction proof for Theorem 2.

- 6) $\mathcal{C}$ picks a random bit $b \leftarrow \{0, 1\}$ and interacts with $\mathcal{B}$ in a run of the anonymous posting protocol, on input $\vec{\sigma}^b$. Simultaneously, $\mathcal{B}$ interacts with $\mathcal{A}$ in the second step of Fusion (where each participant sends their invitation to the bulletin board), on input $\vec{X}^b$. (In these protocols, $\mathcal{A}$ acts on behalf of the corrupted parties, while $\mathcal{C}$ acts on behalf of the honest ones; $\mathcal{B}$ acts as a passive channel to relay the messages between $\mathcal{A}$ and $\mathcal{C}$.)
- 7) $\mathcal{A}$ computes a guess b' for b and then sends it to $\mathcal{B}$. $\mathcal{B}$ outputs the same bit b' and wins the anonymity game if $b' = b$. See Figure 6 for a visual map of the reduction.

Analysis of the reduction. The input vectors $\vec{\sigma}^0, \vec{\sigma}^1$ are equivalent w.r.t. the set of corrupted parties. Concretely, we assume the set Bad contains BB and some participants. Since $\mathcal{A}$ must choose $\vec{X}^0, \vec{X}^1$ such that $\text{PopularityPattern}(\vec{X}^0) = \text{PopularityPattern}(\vec{X}^1)$, the number of invitations for each recipient P_i in $\vec{X}^0$ is the same as that in $\vec{X}^1$. Thus, the set of messages for the bulletin board (the only corrupted party who receives messages in the posting game) is the same, regardless of whether the input is $\vec{X}^0$ or $\vec{X}^1$. In both scenarios, the input for the bulletin board is $\{\}$.

Similarly, since $\mathcal{A}$ must also choose $\vec{X}^0, \vec{X}^1$ such that $\text{InputPattern}(\vec{X}^0) = \text{InputPattern}(\vec{X}^1)$, the inputs of the corrupted participants are the same. In both scenarios, no participant, honest or corrupted, receives any message.

The reduction works because (i) it runs in polynomial time (assuming that Π runs in polynomial time), (ii) the invitations are distributed according to what it would be in Fusion, and, moreover, (iii) $\mathcal{B}$ wins the anonymity game whenever $\mathcal{A}$ wins the dialing game. $\square$

Appendix B. Fusion+: Proof of Fairness

Let Q_j be the total number of invitations to P_j that are not converted into noisy invitations, and $S_{i \rightarrow j}$, the number of those invitations originating from P_i . Hence,

$$S_{i \rightarrow j} \leftarrow \text{Bin}(C_{i \rightarrow j}, 1 - p)$$

$$Q_j = \sum_{i=1}^N S_{i \rightarrow j}$$

Because at most d invitations for P_j can be realized, we have

$$M_{i \rightarrow j} = \begin{cases} S_{i \rightarrow j}, & Q_j \leq d, \\ \frac{S_{i \rightarrow j}}{Q_j} d, & Q_j > d, \end{cases}$$

which is equivalent to $M_{i \rightarrow j} = S_{i \rightarrow j} \min\left(1, \frac{d}{Q_j}\right)$.

Applying the law of total expectation over the two regimes, ($Q_j \leq d$) and ($Q_j > d$), gives:

$$\mathbb{E}[M_{i \rightarrow j}] = \mathbb{E}[M_{i \rightarrow j} \mid Q_j \leq d] \cdot \Pr[Q_j \leq d] + \mathbb{E}[M_{i \rightarrow j} \mid Q_j > d] \cdot \Pr[Q_j > d].$$

When $Q_j \leq d$, all these invitations will be realized, so $M_{i \rightarrow j} = S_{i \rightarrow j}$, and so:

$$\mathbb{E}[M_{i \rightarrow j} \mid Q_j \leq d] = \mathbb{E}[S_{i \rightarrow j} \mid Q_j \leq d] = \frac{C_{i \rightarrow j}}{C_j} \mathbb{E}[Q_j \mid Q_j \leq d].$$

If instead $Q_j > d$, exactly d of these Q_j invitations will be realized, chosen uniformly at random. Conditioning first on $Q_j = s$ yields:

$$\mathbb{E}\left[\frac{d}{Q_j} S_{i \rightarrow j} \mid Q_j = s\right] = \frac{d}{s} \mathbb{E}[S_{i \rightarrow j} \mid Q_j = s] = \frac{d}{s} \frac{C_{i \rightarrow j}}{C_j} s = d \frac{C_{i \rightarrow j}}{C_j}$$

Since this is independent of $s > d$ it follows that

$$\begin{aligned} \mathbb{E}[M_{i \rightarrow j} \mid Q_j > d] &= \mathbb{E}\left[\frac{d}{Q_j} S_{i \rightarrow j} \mid Q_j > d\right] \\ &= \mathbb{E}[\mathbb{E}[M_{i \rightarrow j} \mid Q_j] \mid Q_j > d] \\ &= d \frac{C_{i \rightarrow j}}{C_j} \end{aligned}$$

Putting these pieces together,

$$\begin{aligned} \mathbb{E}[M_{i \rightarrow j}] &= \frac{C_{i \rightarrow j}}{C_j} (\mathbb{E}[Q_j \mid Q_j \leq d] \cdot \Pr[Q_j \leq d] + d \cdot \Pr[Q_j > d]) \\ &= \frac{C_{i \rightarrow j}}{C_j} \mathbb{E}[\min(Q_j, d)] \\ &= \frac{C_{i \rightarrow j}}{C_j} \mathbb{E}[M_j]. \end{aligned}$$

Therefore, we have $\mathbb{E}[M_{i \rightarrow j}] = \frac{C_{i \rightarrow j}}{C_j} \mathbb{E}[M_j]$. So, Fusion+ is fair.

Appendix C. Fusion+: Proof of Differential Privacy

The following argument will refer to the notation in Table 1.

Symbol	Description
$h \in (0, 1]$	Fraction of honest participants in the system
hN	Total number of honest participants
hNd	Total number of invitations sent by honest participants
$Y_{i,j}$	Noisy version of $X_{i,j}$
$\vec{Y} = (Y_1, \dots, Y_N)$	Full noisy input vector after randomization
$\vec{Y}_H$	Subset of $\vec{Y}$ from honest participants only
$\vec{X}_C$	Inputs of corrupted (non-honest) participants
$\vec{b}_H$	Binary indicator vector for whether each honest invitation is “noised”
C_j^{H-}	Number of noised invitations targeting participant P_j , from honest parties
$c = \mathbb{E}[C_j^{H-}]$	Expected number of noisy invitations to P_j
$\text{View}^{\mathcal{A}}(\vec{X})$	Adversary’s view: $(\vec{Y}_H, \vec{X}_C, \vec{b}_H)$
ϵ, δ	Differential privacy parameters

TABLE 1: Table of notation.

Proof of Theorem 4. Let $\vec{X}^0$ and $\vec{X}^1$ be neighboring inputs, differing in exactly one element of a single honest participant’s input. Without loss of generality, assume the participant P_N is honest, and that the differing element is $X_{N,d}$. We write

$$\begin{aligned} \vec{X}^0 &= (X_1^0, X_2^0, \dots, X_N^0) = \begin{pmatrix} \{X_{1,1}, \dots, X_{1,d}\}, \\ \{X_{2,1}, \dots, X_{2,d}\}, \\ \dots \\ \{X_{N,1}, \dots, X_{N,d}\} \end{pmatrix} \\ \vec{X}^1 &= (X_1^1, X_2^1, \dots, X_N^1) = \begin{pmatrix} \{X_{1,1}, \dots, X_{1,d}\}, \\ \{X_{2,1}, \dots, X_{2,d}\}, \\ \dots \\ \{X_{N,1}, \dots, X'_{N,d}\} \end{pmatrix}, \end{aligned}$$

where $X_i^0 = X_i^1$ for all $i < N$, and the only discrepancy is in the last coordinate of P_N .

Next, let $\vec{Y} = (Y_1, \dots, Y_N)$ denote the “noised” input after applying the differential privacy mechanism. Each $Y_i = \{Y_{i,1}, \dots, Y_{i,d}\}$ is generated by replacing each original invitee $X_{i,j}$ with

$$Y_{i,j} = \begin{cases} X_{i,j}, & \text{with probability } 1 - p, \\ X'_{i,j}, & \text{with probability } p. \end{cases}$$

where $X'_{i,j} \leftarrow \mathcal{P}$.

If no noise is added or if the random sample happens to equal the original invitee, then $Y_{i,j} = X_{i,j}$.

We assume the adversary view consists of:

- 1) The noised inputs $\vec{Y} = (Y_1, \dots, Y_N)$ from all parties. Let $\vec{Y}_H$ denote its restriction to honest parties.
- 2) The inputs $\vec{X}_C$ of all corrupted participants.
- 3) A binary indicator vector

$$\vec{b} = (b_{1,1}, b_{1,2}, \dots, b_{1,d}, \dots, b_{N,1}, \dots, b_{N,d}),$$

where

$$b_{i,j} = \begin{cases} 1, & \text{if the invitation corresponding to } X_{i,j} \text{ is real,} \\ 0, & \text{otherwise.} \end{cases}$$

Let $\vec{b}_H$ denote its restriction to honest parties.

Hence, the adversary $\mathcal{A}$'s view on input $\vec{X}$ is $\text{View}^{\mathcal{A}}(\vec{X}) = (\vec{Y}, \vec{X}_C, \vec{b})$.

To show that $\text{View}^{\mathcal{A}}(\vec{X})$ satisfies (ϵ, δ) -DP, it suffices to prove

$$\Pr_{V \leftarrow \text{View}^{\mathcal{A}}(\vec{X})} \left[\frac{\Pr[\text{View}^{\mathcal{A}}(\vec{X}^0) = V]}{\Pr[\text{View}^{\mathcal{A}}(\vec{X}^1) = V]} \geq e^\epsilon \right] \leq \delta.$$

Fix an arbitrary V in the support of $\text{View}^{\mathcal{A}}$ and condition on $\text{View}^{\mathcal{A}}(\cdot) = V$.

$$\text{Let } p_0 = \Pr[\text{View}^{\mathcal{A}}(\vec{X}^0) = V] \text{ and } p_1 = \Pr[\text{View}^{\mathcal{A}}(\vec{X}^1) = V]$$

First, we can consider the case $b_{N,d} = 0$. Here the invitation for $X_{N,d}$ is a noise invitee, independent of $X_{N,d}$. Thus

$$p_0 = p_1,$$

and the privacy bound holds immediately.

Now, let's assume that $b_{N,d} = 0$ when the d -th invitation of P_N is unnoised. Let f be the number of honest invitations that were replaced by noise, and for each $j \in [N]$, let C_j^{H-} denote the count of noised invitations targeting P_j among honest parties, including the real (unnoised) invitation corresponding to $X_{N,d}$. Without loss of generality, suppose $X_{N,d} = P_1$ and $X'_{N,d} = P_2$. Then

$$p_0 = \left(C_1^{H-} - 1, C_2^{H-}, \dots, C_N^{H-} \right) \frac{p^f (1-p)^{hNd-f}}{N^f},$$

and similarly,

$$p_1 = \left(C_1^{H-}, C_2^{H-} - 1, \dots, C_N^{H-} \right) \frac{p^f (1-p)^{hNd-f}}{N^f}.$$

The first factor captures the probability of exactly that specific pattern of noise injections among the honest invitations (i.e. which invitations were replaced by noise and which remained genuine), while the second factor captures the probability that, once those noisy invitations each pick a recipient uniformly at random from the N parties, they yield precisely the observed PopularityPattern histogram.

Therefore, taking the ratio between the last two probabilities we find that, in the case $b_{N,d} = 1$,

$$\frac{p_0}{p_1} = \frac{C_1^{H-}}{C_2^{H-}}.$$

Now note that for $V \leftarrow \text{View}^{\mathcal{A}}(\vec{X})$ the count C_1^{H-} follows a binomial distribution $\text{Bin}(hNd - 1, p/N)$, and $C_2^{H-} - 1$ follows the same distribution. Thus, we have

$$\Pr_{V \leftarrow \text{View}^{\mathcal{A}}(\vec{X})} \left[\frac{p_0}{p_1} \geq e^\epsilon \right] = \Pr \left[\frac{C_1^{H-}}{C_2^{H-}} \geq e^\epsilon \right],$$

where $C_1^{H-} \leftarrow \text{Bin}(hNd - 1, p/N) + 1$ and $C_2^{H-} \leftarrow \text{Bin}(hNd - 1, p/N)$.

We now bound the probability above using a union bound and the multiplicative Chernoff bound. Let

$$c = \mathbb{E}[C_2^{H-}] = \frac{p(hNd - 1)}{N}.$$

Since $\frac{C_1^{H-}}{C_2^{H-}} \geq e^\epsilon$ implies that either $C_1^{H-} \geq ce^{\epsilon/2}$ or $C_2^{H-} \leq ce^{-\epsilon/2}$, we have

$$\begin{aligned} \Pr \left[\frac{C_1^{H-}}{C_2^{H-}} \geq e^\epsilon \right] &\leq \Pr[C_1^{H-} \geq ce^{\epsilon/2}] + \Pr[C_2^{H-} \leq ce^{-\epsilon/2}] \\ &= \Pr[C_2^{H-} \geq ce^{\epsilon/2} - 1] \\ &\quad + \Pr[C_2^{H-} \leq ce^{-\epsilon/2}] \\ &= \Pr \left[C_2^{H-} - c \geq c(e^{\epsilon/2} - 1 - \frac{1}{c}) \right] \\ &\quad + \Pr[C_2^{H-} - c \leq c(e^{-\epsilon/2} - 1)] \end{aligned}$$

Applying the Chernoff bound to each term yields

$$\begin{aligned} \Pr \left[\frac{C_1^{H-}}{C_2^{H-}} \geq e^\epsilon \right] &\leq \exp \left(-\frac{c}{3} (e^{\epsilon/2} - 1 - \frac{1}{c})^2 \right) \\ &\quad + \exp \left(-\frac{c}{2} (1 - e^{-\epsilon/2})^2 \right) \end{aligned}$$

To ensure that $\Pr \left[\frac{C_1^{H-}}{C_2^{H-}} \geq e^\epsilon \right] \leq \delta$, it suffices to require that each exponential term is at most $\delta/2$. Equivalently, we impose

$$\exp\left(-\frac{c}{3}\left(e^{\epsilon/2} - 1 - \frac{1}{c}\right)^2\right) \leq \frac{\delta}{2},$$

$$\exp\left(-\frac{c}{2}\left(1 - e^{-\epsilon/2}\right)^2\right) \leq \frac{\delta}{2}.$$

Taking logarithms then yields

$$c \geq \max\left\{\frac{3\log(2/\delta)}{\left(e^{\epsilon/2} - 1 - \frac{1}{c}\right)^2}, \frac{2\log(2/\delta)}{\left(1 - e^{-\epsilon/2}\right)^2}\right\}.$$

Remark 4. Here we derive Corollary 2

Recalling that $c = \frac{p(hNd-1)}{N}$, we conclude:

$$p \geq \max\left\{\frac{3N\log(2/\delta)}{(hNd-1)\left(e^{\epsilon/2} - 1 - \frac{N}{p(hNd-1)}\right)^2}, \frac{2N\log(2/\delta)}{(hNd-1)(1 - e^{-\epsilon/2})^2}\right\}$$

In other words,

$$c \geq \max\left\{\frac{3\log(2/\delta)}{\left(e^{\epsilon/2} - 1 - \frac{1}{c}\right)^2}, \frac{2\log(2/\delta)}{\left(1 - e^{-\epsilon/2}\right)^2}\right\}.$$

For the second term $\frac{2\log(2/\delta)}{\left(1 - e^{-\epsilon/2}\right)^2}$, when $\epsilon \leq 1$,

$$(1 - e^{-\frac{\epsilon}{2}}) \geq (1 - e^{-\frac{1}{2}})\epsilon \geq \epsilon\sqrt{7}$$

Hence,

$$\frac{2\log(2/\delta)}{\left(1 - e^{-\epsilon/2}\right)^2} \leq \frac{14\log(2/\delta)}{\epsilon^2}$$

For the first term $\frac{3\log(2/\delta)}{\left(e^{\epsilon/2} - 1 - \frac{1}{c}\right)^2}$, if $c \geq \frac{27}{\epsilon}$, then

$$\left(e^{\epsilon/2} - 1 - \frac{1}{c}\right) \geq \left(\frac{\epsilon}{2} - \frac{\epsilon}{27}\right) = \frac{25\epsilon}{54}$$

Therefore,

$$\frac{3\log(2/\delta)}{\left(e^{\epsilon/2} - 1 - \frac{1}{c}\right)^2} \leq \frac{3\log(2/\delta)}{\frac{25^2}{54^2}\epsilon^2} \leq \frac{14\log(2/\delta)}{\epsilon^2}$$

Combining both bounds gives

$$c \geq \max\left\{\frac{14\log(2/\delta)}{\epsilon^2}, \frac{27}{\epsilon}\right\}.$$

Recalling that $c = \frac{p(hNd-1)}{N}$, we finally obtain

$$p \geq \max\left\{\frac{14N\log(2/\delta)}{(hNd-1)\epsilon^2}, \frac{27N}{(hNd-1)\epsilon}\right\}$$

□